

Amir (Ghasem) Boshrouei Shargh

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PhD researcher applying machine learning and uncertainty quantification to develop physics-informed models for structural engineering problems, with a focus on Bayesian optimization and active learning.

Education

PhD in Architectural Engineering
Expertise in Structural Engineering

Jan. 2023 – Present
(NSF-funded)

The Pennsylvania State University, PA, USA

CGPA (WES): 3.8/4

NHERI Converging Design Project

Multi-objective Optimization of Resilient Structural Spines

ICDS Mid-Scale Seed Grant

Reinforcement Learning for High-Dimensional Adaptive Surrogate Modeling

Relevant Courses:

Probabilistic Structural Reliability Analysis, Computational Intelligence and Analysis of Randomness, Stochastic Processes and Monte Carlo Methods, Pattern Recognition and Machine Learning, Structural Design for Dynamic Loads, Computational Design and Optimization.

MSc in Civil & Environmental Engineering
Expertise in Earthquake Engineering

Sep. 2015 – Sep. 2018
(Full Scholarship)

Ferdowsi University of Mashhad, Iran

CGPA (WES): 3.88/4

Thesis Title: *Seismic Assessment of Structures Considering Spectral Shape Parameters*

Relevant Courses:

Random Vibrations, Seismic Structural Design, Probabilistic Seismic Hazard Analysis, Advanced Seismology, Structural Rehabilitation and Retrofitting, Structural Dynamics, Theory of Elasticity and Plasticity.

BSc in Civil & Environmental Engineering

Sep. 2011 – Jul. 2015
(Full Scholarship)

Shahrood University of Technology, Iran

CGPA (WES): 3.13/4

Publications

- **Shargh, G. B.**, Barbosa, A. R., van de Lindt, J. W., Brown, N. C. (2026) “A Parametric Computational Framework for Modeling and Seismic Design of Post-Tensioned Mass Timber Rocking Walls.” To be submitted.
- **Shargh, G. B.**, Brown, N. (2026) “Physics-Aware Bayesian Optimization for Reliable PT-CLT Rocking Walls: Impact of Feature Representations.” Accepted for oral presentation at the 13th U.S. National Conference on Earthquake Engineering (13NCEE), Earthquake Engineering Research Institute (EERI).
- **Shargh, G. B.**, Papakonstantinou, K. G., van de Lindt, J., Barbosa, A., Simpson, B., Sinha, A., Brown, N. (2026) “Reliability-Based Design Optimization of Mass Timber Self-Centering Rocking Walls.” *Journal of Structural Engineering*, In-press.
- **Shargh, G. B.**, Barbosa, A., Simpson, B., Sinha, A., van de Lindt, J. W., Brown, N. (2024) “Multi-Objective Optimization of Cross-Laminated Timber Shear Walls.” 18th World Conference on Earthquake Engineering.
- **Shargh, G. B.**, Barati, R., Brown, N. (2024) “Angular Efficiency of Intensity Measures.” 18th World Conference on Earthquake Engineering.

- Barati, R., **Shargh, G. B.** (2024) “*Seismic Intensity Measure Selection Considering Record-to-Record and Angle-to-Angle Uncertainties.*” 9th International Conference on Seismology and Earthquake Engineering.
- **Shargh, G. B.**, Barati, R. (2023) “*Probabilistic Assessment of Ground Motion Rotation Impact on Seismic Performance of Structures.*” *Engineering Structures*. DOI: 10.1016/j.engstruct.2023.116383.
- **Shargh, G. B.**, Barati, R. (2021) “*Estimation of Inelastic Seismic Input Energy.*” *Soil Dynamics and Earthquake Engineering*. DOI: 10.1016/j.soildyn.2020.106505.
- Barati, R., **Shargh, G. B.**, Karamodin, A. (2019) “*Evaluation of Methods Estimating Structural Strength Reduction Factor under Far-Field Records.*” 8th International Conference on Seismology & Earthquake Engineering.

Open-Source Software

- [Parametric-PT-CLT](#) — A Python-based parametric finite element framework for modeling, optimization, and probabilistic assessment of post-tensioned cross-laminated timber (PT-CLT) rocking wall systems. Supports parametric design space generation, uncertainty quantification workflows, and integration with ML-based surrogate modeling.

Professional Experience

Research Assistant, Pennsylvania State University Jan. 2023 – Present

- Developed Python-based parametric FE and probabilistic models for PT-CLT rocking walls.
- Performed UQ, sensitivity analysis, and physics-aware Bayesian optimization.
- Integrated ML-based surrogate modeling for efficient multi-objective design exploration and robustness assessment.

Research Assistant, Ferdowsi University of Mashhad Oct. 2018 – Jan. 2023

- Conducted probabilistic seismic analysis and nonlinear dynamic simulations using OpenSees.

Mathematics and Statistics Tutor Sep. 2015 – Jul. 2019

- Taught mathematics and statistics through private tutoring and institutional courses.

Awards and Honors

- Awarded a Midscale-Funded Project grant from the Institute for Computational and Data Sciences (ICDS), Pennsylvania State University.
- Awarded a full Research Assistantship (RA) for an NSF-funded research project.
- Recipient of full national scholarships for both graduate and undergraduate studies.
- Graduated in the top 5% of the M.Sc. class of 2018.
- Ranked in the top 4% nationwide in the Iranian M.Sc. entrance exam for Civil Engineering.
- Ranked in the top 0.9% among 300,000 participants in the Iranian national university entrance exam.
- Achieved 6th place in the Khwarizmi National RoboCup Student Competition for developing autonomous agents in C++.

Technical Skills

Programming & ML Frameworks: Python (NumPy, SciPy, TensorFlow, PyTorch), MATLAB, R, C++, Fortran

Engineering & Simulation Software: OpenSees, ETABS, SAP2000, AutoCAD

Specialized Competencies: Probabilistic Structural Analysis, Uncertainty Quantification, Reliability-Based Design, Bayesian Optimization, Surrogate Modeling, Reinforcement Learning, Nonlinear Dynamic Analysis, Seismic Risk Assessment